



AI SOFTWARE DEVELOPMENT INTERNSHIP

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Week 6 — Detailed Task Sheet

LangChain & LCEL • Advanced Retrieval • Tool Calling Resilient .NET fi AI
Service Integration • End-to-End Streaming • Git Rebase

PART B — DOCUMENT LOADERS, SMARTER SPLITTING & ADVANCED RETRIEVAL

Outputs

```
(venv) PS D:\Week6\Week6_PartB_Advanced_Retrieval> python advanced_retrieval.py
=====
PART B - RECURSIVE CHARACTER TEXT SPLITTER
=====
Original corpus records: 1
Generated chunks: 1

--- Chunk 1 ---
Book Title: The Alchemists Updated. Author ID: 1. Category ID: 1.

Recursive splitting completed successfully.
(venv) PS D:\Week6\Week6_PartB_Advanced_Retrieval> █
```

```
● (venv) PS D:\Week6\Week6_PartB_Advanced_Retrieval> python plain_retriever.py
=====
PART B - PLAIN RETRIEVER
=====
Original corpus records: 1
Generated chunks: 1

QUESTION:
What is the title of the book in the library?

RETRIEVED DOCUMENTS:
-----

Result 1:
Book Title: The Alchemists Updated. Author ID: 1. Category ID: 1.
Metadata: {'chunk_id': 1, 'source': 'week5_books_corpus'}

Plain retriever completed successfully.
○ (venv) PS D:\Week6\Week6_PartB_Advanced_Retrieval> █
```

```

• (venv) PS D:\Week6\Week6_Part8_Advanced_Retrieval> python comparison.py
=====
PART B - PLAIN RETRIEVER VS MULTIQUERY RETRIEVER
=====

=====
QUESTION 1
=====
Which planet is third from the Sun?

--- PLAIN RETRIEVER RESULTS ---
Result 1:
Book Title: The Alchemists Updated. Author ID: 1. Category ID: 1.

--- MULTIQUERY RETRIEVER RESULTS ---
D:\Week6\Week6_Part8_Advanced_Retrieval\venv\Lib\site-packages\langchain_google_genai\chat_models.py:3988: UserWarning: Model 'gemini-3.5-flash-lite' uses fixed sampling defaults; the sampling parameter
perature will be ignored.
  request = self._build_request_config(
WARNING:google_genai.models:Direct use of automatic function calling (AFC) in Models.generate_content is not recommended. Instead, we recommend to use AFC in Chat.send_message. Similarly, direct use of
Models.generate_content_stream is not recommended. Instead, we recommend to use AFC in Chat.send_message_stream.
INFO:langchain_classic.retrievers.multi_query:Generated queries: ['What is the third planet from the Sun?', 'Name the planet that is the third closest to the Sun in our solar system.', 'Which planet com
d in order of distance from the Sun?']
Result 1:
Book Title: The Alchemists Updated. Author ID: 1. Category ID: 1.

=====
QUESTION 2
=====
What programming language is used for web development, automation, and artificial intelligence?

--- PLAIN RETRIEVER RESULTS ---
Result 1:
Book Title: The Alchemists Updated. Author ID: 1. Category ID: 1.

--- MULTIQUERY RETRIEVER RESULTS ---
D:\Week6\Week6_Part8_Advanced_Retrieval\venv\Lib\site-packages\langchain_google_genai\chat_models.py:3988: UserWarning: Model 'gemini-3.5-flash-lite' uses fixed sampling defaults; the sampling parameter
perature will be ignored.
  request = self._build_request_config(
INFO:langchain_classic.retrievers.multi_query:Generated queries: ['Which programming language is commonly used in web development, task automation, and AI?', 'What versatile coding language powers web a
ions, automation scripts, and machine learning?', 'Name the language frequently used for building websites, automating processes, and artificial intelligence projects.'].
Result 1:
Book Title: The Alchemists Updated. Author ID: 1. Category ID: 1.

=====
QUESTION 3
=====

```

```

=====
QUESTION 3
=====
Where is the Amazon rainforest located mainly?

(venv) PS D:\Week6\Week6_Part8_Advanced_Retrieval> python comparison.py
Result 1:
Book Title: The Alchemists Updated. Author ID: 1. Category ID: 1.

--- MULTIQUERY RETRIEVER RESULTS ---
D:\Week6\Week6_Part8_Advanced_Retrieval\venv\Lib\site-packages\langchain_google_genai\chat_models.py:3988: UserWarning: Model 'gemini-3.5-flash-lite' uses fixed sampling defaults; the sampling parameter(s) temperature will be ignored.
  request = self._build_request_config(
INFO:langchain_classic.retrievers.multi_query:Generated queries: ['Which countries contain the largest part of the Amazon rainforest?', 'What is the primary geographic region and country of the Amazon r
ainforest lie?']
Result 1:
Book Title: The Alchemists Updated. Author ID: 1. Category ID: 1.

=====
QUESTION 4
=====
What skills are commonly used in basketball?

--- PLAIN RETRIEVER RESULTS ---
Result 1:
Book Title: The Alchemists Updated. Author ID: 1. Category ID: 1.

--- MULTIQUERY RETRIEVER RESULTS ---
D:\Week6\Week6_Part8_Advanced_Retrieval\venv\Lib\site-packages\langchain_google_genai\chat_models.py:3988: UserWarning: Model 'gemini-3.5-flash-lite' uses fixed sampling defaults; the sampling parameter(s) temperature will be ignored.
  request = self._build_request_config(
INFO:langchain_classic.retrievers.multi_query:Generated queries: ['What are the essential techniques and abilities required to play basketball?', 'Which physical and tactical skills are most important for basketball players?', 'What fundamental skills do
you need to master to be good at basketball?']
Result 1:
Book Title: The Alchemists Updated. Author ID: 1. Category ID: 1.

=====
QUESTION 5
=====
What is the capital of Japan?

--- PLAIN RETRIEVER RESULTS ---
Result 1:
Book Title: The Alchemists Updated. Author ID: 1. Category ID: 1.

--- MULTIQUERY RETRIEVER RESULTS ---
D:\Week6\Week6_Part8_Advanced_Retrieval\venv\Lib\site-packages\langchain_google_genai\chat_models.py:3988: UserWarning: Model 'gemini-3.5-flash-lite' uses fixed sampling defaults; the sampling parameter(s) temperature will be ignored.
  request = self._build_request_config(
INFO:langchain_classic.retrievers.multi_query:Generated queries: ['What city serves as the capital of Japan?', 'Name the capital city of Japan.', 'Japan's capital is which city?']
Result 1:
Book Title: The Alchemists Updated. Author ID: 1. Category ID: 1.

=====
5-QUESTION COMPARISON COMPLETED
=====

```

```
(venv) PS D:\Week6\Week6_PartB_Advanced_Retrieval> python multi_query_retriever.py

GENERATING ALTERNATIVE QUERIES...
The generated queries will appear below through logging.
D:\Week6\Week6_PartB_Advanced_Retrieval\venv\Lib\site-packages\langchain_google_genai\chat_models.py:3908: UserWarning: Model 'gemini-3.5-f
lash-lite' uses fixed sampling defaults; the sampling parameter(s) temperature will be ignored.
    request = self._build_request_config(
WARNING:google_genai.models:Direct use of automatic function calling (AFC) in Models.generate_content is not recommended. Instead, we recom
mend to use AFC in Chat.send_message. Similarly, direct use of AFC in Models.generate_content_stream is not recommended. Instead, we recomm
end to use AFC in Chat.send_message_stream.
INFO:langchain_classic.retrievers.multi_query:Generated queries: ['1. Can you tell me the name of the book available in the library?', '2.
What is the book's title located in this library collection?', '3. Could you provide the title of the book housed in the library?']

RETRIEVED DOCUMENTS:
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Result 1:
Book Title: The Alchemists Updated. Author ID: 1. Category ID: 1.
Metadata: {'chunk_id': 1, 'source': 'week5_books_corpus'}

MultiQueryRetriever completed successfully.
(venv) PS D:\Week6\Week6_PartB_Advanced_Retrieval>
(venv) PS D:\Week6\Week6_PartB_Advanced_Retrieval>
(venv) PS D:\Week6\Week6_PartB_Advanced_Retrieval>
```

```
(venv) PS D:\Week6\Week6_PartB_Advanced_Retrieval> python multiquery_worse_case.py

QUESTION:
Which language is useful for programming?

--- PLAIN RETRIEVER ---
INFO:httpx:HTTP Request: POST https://generativelanguage.googleapis.com/v1beta/models/gemini-embedding-001:batchEmbedContents "HTTP/1.1 200 OK"
Result 1:
Python is widely used for web development, automation, and artificial intelligence.

--- MULTIQUERY RETRIEVER ---
D:\Week6\Week6_PartB_Advanced_Retrieval\venv\Lib\site-packages\langchain_google_genai\chat_models.py:3908: UserWarning: Model 'gemini-3.5-flash-lite' uses fi
d sampling defaults; the sampling parameter(s) temperature will be ignored.
    request = self._build_request_config(
INFO:google_genai.models:AFC is enabled with max remote calls: 10.
WARNING:google_genai.models:Direct use of automatic function calling (AFC) in Models.generate_content is not recommended. Instead, we recommend to use AFC in
hat.send_message. Similarly, direct use of AFC in Models.generate_content_stream is not recommended. Instead, we recommend to use AFC in Chat.send_message_st
am.
INFO:httpx:HTTP Request: POST https://generativelanguage.googleapis.com/v1beta/models/gemini-3.5-flash-lite:generateContent "HTTP/1.1 200 OK"
INFO:langchain_classic.retrievers.multi_query:Generated queries: ['What are the most popular and practical programming languages to learn?', 'Which coding la
uages should I study for software development?', 'What are the best programming languages for different tech careers?']
INFO:httpx:HTTP Request: POST https://generativelanguage.googleapis.com/v1beta/models/gemini-embedding-001:batchEmbedContents "HTTP/1.1 200 OK"
INFO:httpx:HTTP Request: POST https://generativelanguage.googleapis.com/v1beta/models/gemini-embedding-001:batchEmbedContents "HTTP/1.1 200 OK"
INFO:httpx:HTTP Request: POST https://generativelanguage.googleapis.com/v1beta/models/gemini-embedding-001:batchEmbedContents "HTTP/1.1 200 OK"
Result 1:
Python is widely used for web development, automation, and artificial intelligence.
Result 2:
Java is used for enterprise software and Android application development.

--- TRADE-OFF ANALYSIS ---
Plain retrieval performs one similarity search.
MultiQueryRetriever generates multiple alternative queries.
The additional queries can improve recall.
However, broader alternative queries can also retrieve documents
that are less relevant to the original question.
Therefore, MultiQueryRetriever trades extra LLM calls and possible
noise for potentially better retrieval coverage.

MultiQuery worse-case challenge completed.
```